# 清华大学热能工程系 2010 级“燃烧学”期中考试试卷（A 卷）

# Middle Term Examination of Combustion (A)

**考试时间(Date)：　2013 年 4 月 26 日　班级(Class)：____**

**学号(Student No.)：_____ 姓名(Name)：_____**

**Instructions(提示):**

1　请将你的学生证放在桌子的右上角（Please put your Photo student ID on the desk）

2　考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）

3　试卷共有8题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有6页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are 11 questions on this exam with 5 pages. You have up to 120 minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4　如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5　强烈推荐你先读一下所有的问题，并对回答问题的时间进行按排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6　请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7　答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

1. 化学反应净生成率 $w_i$ 的理解(10 分)

(1) 某组分 *i* 净生成率 $w_i$ 单位是 kmol/m$^3$-s, 你认为其定义式 $w_i = \frac{d[N_i/V]}{dt}$ 正确吗？并给出解释（4 分）。

(2) 全混流反应器认为反应器内返混无穷大，即扩散十分快，从而导致反应器内各点和出口的参数性质一致，而组分守恒方程却可写成 $w_i + \frac{\dot{m}}{\rho V}([X_i]_{in} - [X_i]_{out}) = 0$，即反应器进、出口化学反应项和对流项的平衡，反而并没有含扩散项，您是如何理解的？(6 points)

(1) $w_i$, of species i, has a unit of kmol/m$^3$-s, do you think whether the definition of $w_i = \frac{d[N_i/V]}{dt}$ is right or not？ Please explain it（4 points）。

(2) The well-stirred reactor has the ideally perfect mixing in the control volume, implying the diffusion is greatly larger than the convection in the reactor. So, the composition of species within the reactor is everywhere the same, while the composition at the outlet must be the same as in the interior. The conservation equation of species *i* can be expressed as $w_i + \frac{\dot{m}}{\rho V}([X_i]_{in} - [X_i]_{out}) = 0$. Note that, **there is no diffusion term, the larger term, in the above equation**. How can you understand it? And explain it. (6 points)

2. 绝热燃烧温度的计算(18 分)

1) 假设产物中没有解离发生，计算下面公式所示的化学反应在 298K，一个大气压情况下的绝热火焰温度。

$$CH_4 + 13(0.21O_2 + 0.79N_2) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 + n_{O_2}O_2$$

2) 将反应物中的氮气 $N_2$ 替换为 $CO_2$，并再次计算绝热火焰温度并分析变化愿意和规律。

The prediction of adiabatic combustion temperature (15 points)

1） Assuming no dissociation of product species, determine the adiabatic flame temperature for the following chemical reaction at $T_i = 298k$, $P$=1atm

$$CH_4 + 13(0.21O_2 + 0.79N_2) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 + n_{O_2}O_2$$

2） Change the $N_2$ to $CO_2$. Determine the adiabatic flame temperature again and analyze the changing trends.

3. 有一个有扩散火焰形成的一维高温反应器，气流速度为 2.5m/s，反应器截面直径为 50mm，反应器温度为 1200K，其中 $D=2.2\times10^{-5}(273K,1atm)$，$Sc=\frac{\nu}{D}=0.8$，计算其贝克利数 $Pe=\frac{\nu L}{D}$ 和雷诺数，并通过贝克利数判断这个反应器是否是柱塞流反应器(10 分)。

A non premixed flame reactor was a one dimensional flow reactor. The gas velocity is 2.5 m/s. The cross section diameter of the reactor is 50 mm. $D=2.2\times10^{-5}(273K,1atm)$, $Sc=\frac{\nu}{D}=0.8$. Calculate the Pe number $Pe=\frac{\nu L}{D}$ and Re number, and analyze whether the reactor is a plug flow reactor.(10 points).

4. 燃料液滴的蒸发对许多实际装置具有重要意义。考虑一个直径 100μm 的正癸烷（$C_{10}H_{22}$）液滴在 800K，1atm 的干燥氮气中蒸发的情形。液滴表面温度为 25℃。用下式

$$K=\frac{8\bar{\rho}D_{C_{10}H_{22},N_2}}{\rho_l}\frac{1-Y_{C_{10}H_{22},\infty}}{1-Y_{C_{10}H_{22},s}}$$

计算**此刻**的蒸发速率常数 K 是否合理（即假设液滴表面温度恒定）？试说明原因。

上式中，$\rho_l$ 是液滴密度，$\bar{\rho}$ 是合理近似下的气体密度，$D_{C_{10}H_{22},N_2}$ 是正癸烷在氮气中的二元扩散系数，$Y_{C_{10}H_{22},s}$、$Y_{C_{10}H_{22},\infty}$ 分别为液滴表面和无穷远处的正癸烷蒸汽质量分数。取 $Y_{C_{10}H_{22},\infty}=0$。计算可能用到的参数见下表。(12 分)

| 燃料<br>fuel | MW<br>（kg/kmol） | $\rho_l$ （kg/m³） | 沸点<br>boiling point<br>（℃） | $h_{fg}$<br>（kJ/kg） |
|---|---|---|---|---|
| 正癸烷<br>n-Decane | 142.284 | ~730 | 174.1 | 277 |

The evaporation of fuel droplet is important for many practical device. Here we consider a droplet of n-decane (C10H22) with a diameter of 100μm surrounded by dry nitrogen atmosphere (800K , 1atm). The temperature of droplet surface is 25 ° C. Would it be reasonable to calculate the evaporation rate constant K using

$$K=\frac{8\bar{\rho}D_{C_{10}H_{22},N_2}}{\rho_l}\frac{1-Y_{C_{10}H_{22},\infty}}{1-Y_{C_{10}H_{22},s}}$$ ?

The surface temperature is assumed to remain unchanged. Please explain why.

$\rho_l$ is the density of droplet, $\bar{\rho}$ is the density of gas phase with appropriate assumptions.

$D_{C_{10}H_{22},N_2}$ is the binary diffusion coefficient of n-decane and nitrogen. $Y_{C_{10}H_{22},s}$ and $Y_{C_{10}H_{22},\infty}$ is

the mass fraction of n-decane gas near the droplet surface and at infinity, respectively. Assume that $Y_{C_{10}H_{22},\infty}=0$. The possibly used physical properties are listed in the table.

5. 将氮氧化物生成的热力型（或泽尔多维奇，Zeldovich）机理简化为以下一步反应：

$$N_2+O_2 \Leftrightarrow 2NO$$

计算该反应在 1600K、2400K 下的标准平衡常数。根据计算结果，比较热力型 NOxs 生成机理与温度相关的变化。**本题要求各物质的热力学数值只能从下表中取值。(14 分)**

The thermal (or Zeldovich) mechanism of NOx formation is simplified to the one-step reaction here:

$$N_2+O_2 \Leftrightarrow 2NO$$

Please calculate the standard equilibrium constant of the reaction at temperatures of 1600K, 2000K and 2400K. Discuss the role of thermal mechanism during NOx formation according to the results.

**Attention: You can ONLY use the thermodynamical data listed in the table below.**

氮气（N2），标准生成焓=0

Nitrogen (N2), standard enthalpy of formation = 0

| T/K | $\left(\bar{h}^0(T)-\bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 41899 | 244.048 |
| 2400 | 70645 | 258.598 |

氧气（O2），标准生成焓=0

Oxygen (O2), standard enthalpy of formation = 0

| T/K | $\left(\bar{h}^0(T)-\bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 44253 | 260.339 |
| 2400 | 74467 | 275.627 |

一氧化氮（NO），标准生成焓= 90297 kJ/kmol

Nitric oxide (NO), standard enthalpy of formation = 90297 kJ/kmol

| T/K | $\left(\bar{h}^0(T)-\bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 43317 | 264.937 |
| 2400 | 72617 | 279.772 |

6. 研究下列一组反应：(15 分)

$M+Br_2 \rightarrow 2Br+M$ (1)

$Br+H_2 \rightarrow HBr+H$ (2)

$H+Br_2 \rightarrow HBr+Br$ (3)

$H+HBr \rightarrow H_2+Br$ (4)

$Br+Br+M \rightarrow Br_2+M$ (5)

1) 试说明上述反应中，哪些是链的激发反应，哪些是链的传递反应，哪些是链的中断反应（3 分）
2) 写出上述反应的总包反应。（2 分）
3) 试写出 d[HBr]/dt，要求用[$H_2$]和[$Br_2$]表示[提示：假设 Br 和 H 处于稳态条件]。（10 分）

As for the reactions bellows: (15 points)

$M+Br_2 \rightarrow 2Br+M$ (1)

$Br+H_2 \rightarrow HBr+H$ (2)

$H+Br_2 \rightarrow HBr+Br$ (3)

$H+HBr \rightarrow H_2+Br$ (4)

$Br+Br+M \rightarrow Br_2+M$ (5)

(1) Please point out the chain-initiation, chain-propagating and chain-terminating reactions.
(2) Write the global reaction rate of above reactions
(3) Write the expression of d[HBr]/dt, only in terms of [$H_2$] and [$Br_2$] [Hint: assume that Br and H are in steady state situation].

7. 甲醇（$CH_3OH$）被认为是一种很有潜力的含氧燃料，其氧化燃烧应该很复杂，显然要全面描述这一系统是很困难的。你是否可以大致描述一下主要的氧化燃烧的过程（假设是在高温条件下，温度为 1800K。（13 分）

Though methanol （$CH_3OH$） is new fuel containing oxygen and its oxidation should vevry complicated, which causes it very difficult to completely describe all these reaction mechanisms. Please briefly summarize the oxidation mechanism of methanol.

8. 假设控制容积内只有性质均为一致的一类分子 A, 则分子的自由程可以表示为 $\lambda_A = \frac{\bar{v}_A}{\Theta_A} = \frac{1}{\sqrt{2}\pi n_A d_A^2}$，其中 $\bar{v}_A = \sqrt{8k_BT/\pi m_A}$ 是分子平均运动速度，而 $\Theta_A$ 是单个分子与周围分子的碰撞频率(8 分)。

(1) 将控制容积内所有分子的总碰撞频率 $Z_{AA}$ 写成 $Z_{AA} = \frac{1}{2} n_A n_A d_A^2 (\frac{8\pi k_B T}{m_{AA^*}})^{1/2}$，你认为是否合理？

(2) 判断 $\lambda_A$、$\Theta_A$、$Z_{AA}$ 与压力的幂次关系是多少？(提示状态方程)

Assuming that there only uniformly A molecules in the control volume, the molecular free

path can be expressed as $\lambda_A = \frac{\bar{v}_A}{\Theta_A} = \frac{1}{\sqrt{2}\pi n_A d_A^2}$ with the mean molecular velocity of $\bar{v}_A = \sqrt{8k_B T / \pi m_A}$. We know $\Theta_A$ as the collision frequency of a single molecule with his neighboring molecules.

(1) If the total collision frequency of all molecules A in the control volume is expressed as $Z_{AA} = \frac{1}{2} n_A n_A d_A^2 (\frac{8\pi k_B T}{m_{AA^*}})^{1/2}$, please judge it reasonable or not?

(2) Please given the dependence of $\lambda_A, \Theta_A$ and $Z_{AA}$ with the pressure ($\propto \mathrm{P}^n$), respectively. That is, please determine the values of *n* for $\lambda_A, \Theta_A$ and $Z_{AA}$, respectively.